

SNEHA DANDEKAR

International Institute of Information Technology, Bangalore

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Education

International Institute of Information Technology Bangalore

Aug 2024 – Present

Master of Science by Research VLSI

3.58/4

Birla Vishvakarma Mahavidyalaya

June 2018 – May 2022

Bachelor of Technology in Electronics Engineering

8.40/10

Work Experience

InSemi Technologies Services Pvt Ltd

March 2024 – August 2024

Design Engineer

Bangalore, Karnataka

- Worked in Qualcomm as Temp with I/O characterization team.
- Majorly worked in the liberty file generation and validation for custom IPs in lower technology nodes such as 4nm and 3nm (TSMC and Samsung).

Larsen and Tubro - Heavy Engineering

June 2022 – February 2023

Graduate Engineer Trainee

Surat, Gujarat

- Worked in Manufacturing Technologies R&D.
- Worked with a multi-domain team to provide automation in welding technologies.

Projects

Bitmap-Based Sparse Accelerator for Efficient Matrix Multiplication | *Vivado, Innovus*

- Currently designing a row-wise stationary dataflow microarchitecture with kernel fusion for data-reuse and aligning sparsity with the hardware resources.
- Implemented configurable intra-block parallelism to balance throughput and sparse efficiency while avoiding unstructured sparsity overheads.

Hardware Software Co-Optimization for RISC-V | *Vivado, Genus, Innovus*

- Designed and implemented a RISC-V co-processor for efficient Hyperdimensional Computing (HDC) targeting low-power edge systems.
- Achieved $16.44\times$ energy efficiency improvement and $21.57\times$ throughput gain over baseline RISC-V cores (FPGA validation).
- Validated ASIC implementation with 15.89% area savings and 12.33% power reduction while maintaining comparable timing.

CORDIC Architecture Design for Hyperbolic Functions | *Vivado, Genus, Innovus*

- Designed scaling-free folded CORDIC module to determine hyperbolic sine and cosine functions. The design involved a Taylor series approximation with binomial expansion to reduce complexity and latency.
- Achieved a frequency of operation of about 135MHz with around 99% accuracy.

Publications

Mixed-Precision Booth Factored Systolic Array Design for Accelerating Neural Networks

Sneha Dandekar, Bhavana S, Madhav Rao

- Accepted in 27th International Symposium on Quality Electronic Design (ISQED)

Split-Posit Systolic Array Architectures for Resource-efficient Neural Networks

Sneha Dandekar, Arun M, Vaishnavi Sharma, Madhav Rao

- Submitted to International Symposium on Low Power Electronics and Design (ISLPED)

Technical Skills

Languages: Verilog, SystemVerilog (Beginner Proficiency), TCL (Beginner Proficiency), Python

Protocols: UART, SPI, I2C, AXI Stream

Tools: Cadence Genus, Cadence Innovus, Synopsys DesignCompiler, Synopsys PrimeTime, Cadence Liberate

Certifications

- Advanced Diploma in Physical Design | RV VLSI, RV Skills
- Cloud Computing | NPTEL
- Joy of learning in Python | NPTEL

Leadership / Extracurricular

- Core member of The Robotic Society (TRS), an Electronics club in BVM.
- Mountaineering, Have trekked thrice at the height of 14000ft in the lower Himalayas
- Bharatnatyam (Completed six years)